AKASH BHOWMICK

Engineering Lead | TypeScript · Python · AI/LLM & Agentic Systems · AWS

Dhaka, Bangladesh (Remote)
akashnsu2016@gmail.com
linkedin.com/in/akash-bhowmick
Claude Certified Architect · AWS SAA

PROFESSIONAL SUMMARY

Engineering lead with 7+ years building and shipping production platforms in Python (Django, FastAPI, Flask), PostgreSQL and AWS for global streaming, fintech and ecommerce clients. Has led cross-functional teams, owned delivery from concept to production, and trains and mentors new engineers as Craftsmen's internal AWS/DevOps trainer.

Hands-on with AI/LLM delivery: agentic pipelines on AWS Bedrock, RAG systems, LLM-driven video processing, and an agentic Claude Code development workflow with sub-agents, MCP and automated PR review. Anthropic Claude Certified Architect and AWS Solutions Architect certified.

CORE SKILLS

Leadership & Delivery	Team leadership · roadmap-to-milestone breakdown · Agile/Scrum · code review · mentoring & technical training · on-call & incident ownership · client technical discussions
AI / LLM	Claude API · Claude Code (sub-agents, MCP, hooks, skills) · AWS Bedrock · agentic pipelines · RAG (pgvector) · LLM APIs (Claude, Gemini) · AI-assisted development
Backend	Python (7 yrs) · Django / Django REST Framework · FastAPI · Flask · REST API design · auth & integrations · CQRS · microservices
Frontend	React · TypeScript · JavaScript
Database	PostgreSQL · SQL · DynamoDB · schema design
Infrastructure	Docker · Linux · Nginx · CI/CD (Jenkins, GitHub Actions, CodePipeline) · AWS (Lambda, Step Functions, ECS, EC2, RDS, VPC) · Terraform · CDK
Quality	Automated testing · BDD (Python Behave) · end-to-end post-deployment test pipelines

WORK EXPERIENCE

Software Development Engineer II · Craftsmen Software *Mar 2021 – Present*

Engineering Leadership & Enablement

- Design and deliver Craftsmen's **AWS, DevOps and serverless training programme** for new engineers; mentor trainees from onboarding to client-ready.
- Lead a company-wide **AI-native adoption initiative**: certification tiers, a new-joiner AI-readiness scorecard and a maturity ladder for agentic development.
- Own **technical direction** on client projects: architecture, task breakdown, code review, deployments and incident response.

Global OTT Streaming Platform *(Flask · DynamoDB · Lambda · MediaLive · Terraform · Jenkins)*

- Built core services for a live-event orchestration platform serving **millions of concurrent viewers** across multiple regions and accounts.
- Led the **zero-downtime migration** of the full live-streaming stack from Python 2.7 to 3.8, with no production incidents.
- Ran **multi-region CI/CD** with Jenkins and Terraform; added load balancing (NLB, Nginx) and auto-scaling on origin servers.
- Built end-to-end **post-deployment test pipelines** (Python Behave/BDD, Docker, Jenkins); handled on-call and root-cause analysis.
- Developed APIs including an **SCTE-35 ad-insertion signalling service**; ran a full live-streaming proof of concept with the client.

Serverless AI Video Pipeline *(Step Functions · Transcribe · Rekognition · Bedrock · Gemini)*

- Architected a serverless **horizontal-to-vertical video pipeline** with LLM-driven content reframing.
- Designed a serverless migration of **petabyte-scale video archives** from S3 Glacier into a media asset management system.

Live News Scheduling Platform *(CDK · VPC · EC2 · RDS)*

- Designed and delivered a news scheduling and orchestration platform, **fully provisioned in CDK** (TypeScript) with secure VPC networking.

Software Engineer · Circle FinTech Ltd. *Aug 2020 – Feb 2021*

- Built a **banking API integration platform** (Django, Flask) on a TCS core banking system, using CQRS and concurrency handling for high-throughput payments (SPP, Jamuna Bank Just Pay MFS).

Software Engineer (Team Lead) · Codesmith Tech Ltd. *2019 – 2020*

- **Led a cross-functional team** of backend and frontend engineers, designers and business analysts to plan, build and launch a multi-vendor ecommerce platform (Python, Django, AWS).

SELECTED AI & PRODUCT PROJECTS

- **Agentic development pipeline (Claude Code):** spec-to-system workflow where role agents (product, engineering manager, developers, QA) turn a PRD into tickets with acceptance criteria, implementation, tests and BDD/e2e runs. Uses sub-agents, MCP, hooks and an automated PR-review skill.
- **Self-hosted RAG platform:** FastAPI, PostgreSQL + pgvector, with switchable Claude API and local Ollama models.
- **Byudy ecommerce:** Django + React microservices with an AI product-upload pipeline that generates titles, descriptions and infographics for vendors to publish.

CERTIFICATIONS

Claude Certified Architect – Foundations
Anthropic · Credly verified

AWS Certified Solutions Architect – Associate
SAA-C03 · Credly verified

EDUCATION & PUBLICATION

BSc in Computer Science & Engineering
North South University, Dhaka · 2017

Published Research — Springer CSOC 2019
Performance Analysis of RNN Architectures and Classical Statistical Models for Financial Forecasting: A Case Study on Dhaka Stock Exchange Link